RETRO TV
PROJECT

By Haidar Ali Alshowaikh | Class 8E



WHAT IT'S BUILT FROM

Materials Used

- Raspberry Pi
- Arduino Uno
- Potentiometer 10K
- Jumper Wires
- Button
- Joystick Module
- Old Laptop Screen
- HDMI Module

ROADMAP

What's Ahead



01

The Idea

How I got the idea



02

The Problems

What went wrong along the way



03

Solutions

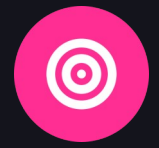
How each problem got fixed



04

The Wiring

How every part connects



05

Goals

What the project set out to do



SECTION 01

THE IDEA

How I got the idea

The Laptop...

I found a laptop that we don't use any more, so I thought of recycling it and giving it a new purpose.



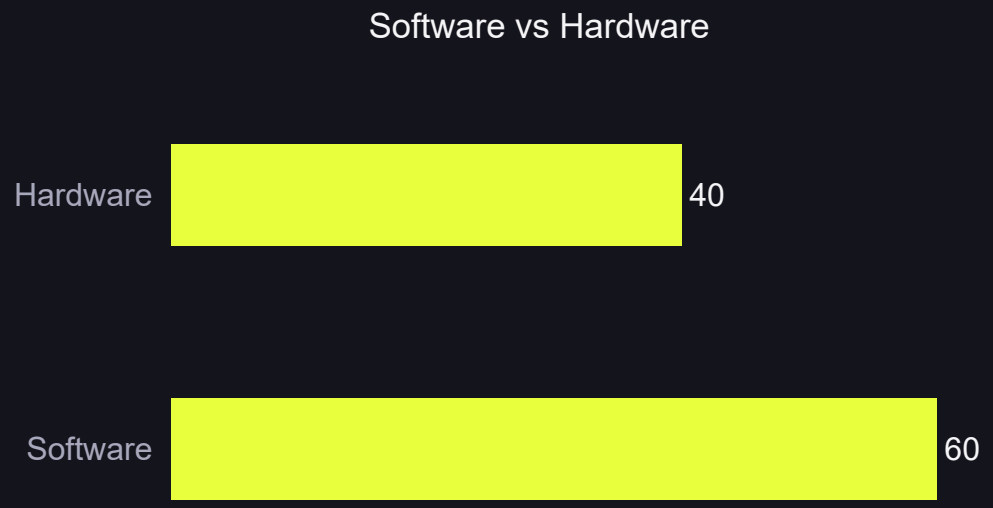


SECTION 02

THE PROBLEMS

What went wrong along the way

Problems By Type



TIME TO FIX

Under a day

Multiple days

Biggest Time Sinks



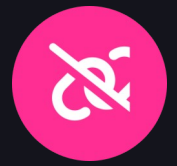
35%

Tracking down the PipeWire bug that sent audio out the wrong HDMI port



40%

Getting the joystick reliably read as arrow keys under the Wayland compositor



25%

Repairing the LCD panel after its LVDS connector latch broke

Component Issues

COMPONENT	ISSUE
Audio output	Routed to the wrong HDMI port
Volume knob	Changed app AND system volume at once
LCD panel	Connector latch broke during repair
Joystick hole	Wider than the module's own screws
Speaker grille	Size was an unconfirmed placeholder
Volume keys	Real multimedia keys grabbed by the desktop

Four Main Problems



Audio Routing

mpv sent sound out the wrong HDMI port



Volume Control

The knob changed app and system volume together



Screen Connector

The connector latch snapped during repair



Hole vs Screws

Joystick hole ended up wider than its screws



SECTION 03

SOLUTIONS

How each problem got fixed

How I Fixed Them



System Volume

Volume knob now calls wpctl to change the real speaker



Audio Device

mpv points straight at the correct ALSA HDMI device



Key Detection

Checks both the keysym and the raw hardware keycode

Debugging Toolbox



wpctl status

Check where audio was actually routing



key_debug.py

Print the real keycode for any key



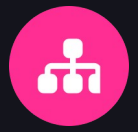
Flashlight test

Tell a dead backlight from dead video



iptv-org

Verify channels before adding them



BFS check

Prove the keyboard is joystick-reachable



Round-trip test

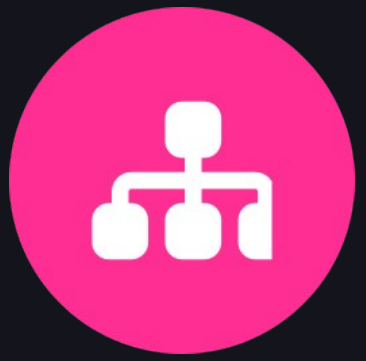
Verify the installer before sending it

Where This Stands Now

Every major software problem has a working fix - audio, volume control, and key detection are all solid.

The screen connector repair is still in progress.





SECTION 04

THE WIRING

How every part connects

Wiring Overview

Joystick

Arduino digital pins, plus one for the click button

Volume Knob

A 10K potentiometer on an Arduino analog input

Power Button

Its own pin, held 2 seconds to shut down

Arduino -> Pi

USB, sending plain text commands over serial

Pi -> Driver Board

HDMI into the universal LCD controller board

Board -> Screen

LVDS ribbon plus a separate backlight connector

How A Button Press Travels



1. ARDUINO

Reads the joystick, knob, and button; sends a command over serial

→



2. THE BRIDGE

rpi_bridge.py turns that command into a real key or mouse event

→

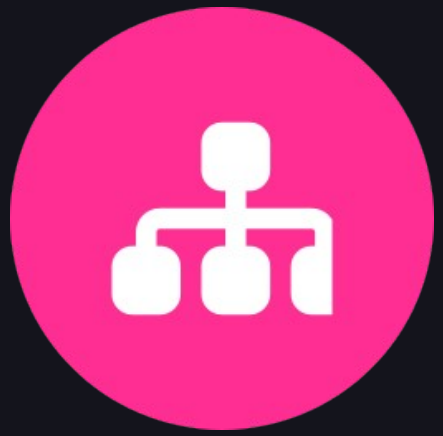


3. THE APP

tv.py receives it like normal keyboard input and reacts on screen

Wiring In Progress

Every input component runs back to the Arduino on its own pin, and the Arduino talks to the Pi over a single USB serial cable.



The Finished Build



Three Main Connections



USB Serial

Arduino to Pi, carrying every joystick, knob, and button command



GPIO

The power button wired straight to the Pi for a direct shutdown



HDMI

Pi to the driver board, carrying video to the recycled screen

Wiring Reference Table

	PORT A	PORT B	PORT C
ARDUINO	Joystick X/Y/click	Knob wiper	Button pin
PI	USB from Arduino	HDMI to driver board	GPIO to power button
DRIVER BOARD	HDMI in	LVDS out to panel	Backlight power out
SCREEN	LVDS in	Backlight in	-
POWER	5V to Pi	12V to driver board	Shared ground

Build Milestones



STEP 1

Wire joystick, knob, and button to the Arduino

→



STEP 2

Flash the Arduino sketch, test raw serial output

→



STEP 3

Get rpi_bridge.py turning serial into key events

→



STEP 4

Wire the driver board, screen, and power button



SECTION 05

GOALS

What the project set out to do

Goals



GOAL 1

Build a fully working retro TV, controllable end-to-end with just the joystick



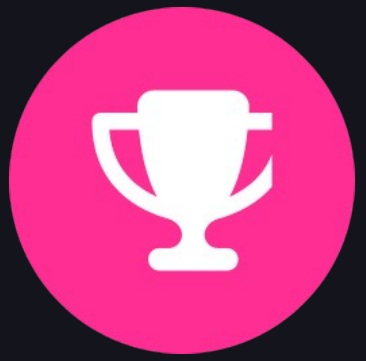
GOAL 2

Fit everything into a custom laser-cut enclosure sized for the recycled screen



264

Channels curated for the custom IPTV channel list, each checked twice before being added



SECTION 06

PROJECT FAIR

How this project fits what judges look for

Why This Fits The Fair

Real Problem Solved

Built a working alternative instead of buying a new TV

Engineering Process

Idea, problems, solutions, wiring, and testing - all documented

Sustainability

Gave a retired laptop screen a second life instead of e-waste

Original Build

Custom Arduino + Pi software, not an off-the-shelf kit

Real-World Impact



Less E-Waste

One fewer laptop thrown out - the screen and body both got reused



Skills, Not Just Parts

Wiring, embedded software, and enclosure design in one build



Built To Show

Everything here actually works, live, not just on paper



THANKS!

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